DAA banana problem 1

Name: Rithvik Rajesh Matta

SRN: PES2UG23CS485

Class: 4H

CODE:

```
#include <stdio.h> #include <limits.h> #include <stdbool.h>

#define MAXN 10

int n; int graph[MAXN][MAXN]; bool visited[MAXN]; int bestPath[MAXN + 1], currentPath[MAXN + 1];
int minCost = INT_MAX;

bool isLexicographicallySmaller(int *path1, int *path2) { for (int i = 0; i < n; i++) { if (path1[i] < path2[i])
return true; if (path1[i] > path2[i]) return false; } return false; }

void dfs(int node, int count, int cost) { if (count == n) { if (graph[node][0] != -1) {

 cost += graph[node][0]; currentPath[n] = 0;

    if (cost < minCost || (cost == minCost && isLexicographicallySmaller(currentPath, bestPath))) {
        minCost = cost;
        for (int i = 0; i <= n; i++)
            bestPath[i] = currentPath[i];
    }
  }
  return;
}

for (int next = 0; next < n; next++) {
  if (!visited[next] && graph[node][next] != -1) {
     visited[next] = true;
     currentPath[count] = next;
     dfs(next, count + 1, cost + graph[node][next]);
     visited[next] = false;
  }
```

```
}

}

int main() { scanf("%d", &n); for (int i = 0; i < n; i++) for (int j = 0; j < n; j++) scanf("%d", &graph[i][j]);

for (int i = 0; i < MAXN; i++)
   visited[i] = false;

visited[0] = true;
currentPath[0] = 0;
dfs(0, 1, 0);

if (minCost == INT_MAX) {
   printf("-1\n");
} else {
   printf("Minimum Cost: %d\n", minCost);
   printf("Optimal Path: ");
   for (int i = 0; i <= n; i++)
      printf("%d ", bestPath[i]);
   printf("\n");
}

return 0;

}
```

OUTPUT:

```
rithvikmatta@penguin:~/sem4/DAA$ gcc bananaProb.c
rithvikmatta@penguin:~/sem4/DAA$ ./a.out
3
0 10 15
10 0 20
15 20 0
Minimum Cost: 45
Optimal Path: 0 1 2 0
```

```
rithvikmatta@penguin:~/sem4/DAA$ gcc bananaProb.c
rithvikmatta@penguin:~/sem4/DAA$ ./a.out
4
0 5 9 10
5 0 6 4
9 6 0 8
10 4 8 0
Minimum Cost: 26
Optimal Path: 0 1 3 2 0
```

```
rithvikmatta@penguin:~/sem4/DAA$ gcc bananaProb.c
rithvikmatta@penguin:~/sem4/DAA$ ./a.out
4
0 -1 -1 4
-1 0 3 -1
-1 3 0 2
4 -1 2 0
The path doesn't exist
```

```
rithvikmatta@penguin:~/sem4/DAA$ gcc bananaProb.c
rithvikmatta@penguin:~/sem4/DAA$ ./a.out
4
0 1 15 6
2 0 7 3
9 6 0 12
10 4 8 0
Minimum Cost: 21
Optimal Path: 0 1 3 2 0
```